Adam Kostka

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Research Profile

I develop methods for making multi-agent LLM systems more reliable, combining statistical calibration, symbolic reasoning and empirical evaluation. As a Computer Science MSc student, I have authored eight published and one accepted paper, with work published at UAI, ICLP and AIED. My research is complemented by experience as an AI Engineer and startup co-founder, building and deploying agent systems.

Education

Warsaw University of Technology — Warsaw, Poland
Ranked 1st among technical universities in Poland (Perspektywy 2026).
MSc in Computer Science, weighted average 4.75/5.00 — Oct 2025 – Expected Jun 2027
BSc in Computer Science, weighted average 4.56/5.00 — Oct 2022 – Jun 2025

Kyungpook National University — Daegu, South Korea
BSc in Computer Science and Engineering — Mar – Dec 2024 (study period)
Double-degree program with Warsaw University of Technology.

Research Experience

Warsaw University of Technology — Warsaw, Poland
Academic research, supervised by Jarosław A. Chudziak — 2024 – Present

- Investigated statistical certification of multi-agent factual outputs using within-response variation in claim-support scores and calibration under a bounded expected false discovery rate [2].
- Studied symbolic-probabilistic belief harmonization under heterogeneous and changing epistemic structures [3,5].
- Designed and evaluated multi-agent systems combining knowledge retrieval, Theory of Mind, memory and logic-based validation across different models and tasks [4,6,7,9]. Applied these approaches to personalized AI tutoring [8].
- Studied runtime governance of agent actions beyond verbal agreement [1].

Publications

1. A. Kostka and J. A. Chudziak, "Beyond Verbal Agreement: Governing Operational Consensus in Agentic AI Systems". HICSS-60, 2027, accepted.
2. A. Kostka and J. A. Chudziak, "Controlling Uncertainty and Hallucination Risk in Multi-Agent Fact Verification". UAI 2026, PMLR 337, pp. 3143–3161 (poster presentation) [paper].
3. A. Kostka and J. A. Chudziak, "Explainable Belief Harmonization under Dynamic Epistemic Partitions". ICLP 2026, Technical Communication, EPTCS 450, pp. 529–543 (presented) [paper].
4. A. Kostka and J. A. Chudziak, "Exploring the Interplay of Theory of Mind, Logic Validation, and Advanced Retrieval in LLM-Based Multi-Agent Systems". Knowledge 6(3), 17, 2026 [paper].
5. A. Kostka and J. A. Chudziak, "Epistemically-Constrained Belief Harmonization: Integrating Symbolic Structure with Probabilistic Consensus". SKILLED-LLMs 2026 (KR workshop), CEUR Workshop Proceedings 4229 (presented) [paper].
6. A. Kostka and J. A. Chudziak, "Evaluating Theory of Mind and Internal Beliefs in LLM-Based Multi-Agent Systems". ICCCI 2025, LNCS 16138, pp. 18–32 [paper].
7. A. Kostka and J. A. Chudziak, "Towards Cognitive Synergy in LLM-Based Multi-Agent Systems: Integrating Theory of Mind and Critical Evaluation". CogSci 2025, pp. 3365–3372 [paper].
8. J. A. Chudziak and A. Kostka, "AI-Powered Math Tutoring: Platform for Personalized and Adaptive Education". AIED 2025, LNCS 15882, pp. 462–469 [paper].
9. A. Kostka and J. A. Chudziak, "Synergizing Logical Reasoning, Knowledge Management and Collaboration in Multi-Agent LLM System". PACLIC 38, 2024, pp. 203–212 (presented) [paper].

Awards and Honours

Senate Scholarship 2026
One of five recipients selected university-wide from students and doctoral candidates at Warsaw University of Technology.

Rector's Scholarship
Awarded three times by Warsaw University of Technology.

Conference Support 2026
UAI Scholarship and ICLP JAI/IJCAI Grant.

Selected Coursework

Completed coursework across the BSc and double-degree studies.

Mathematics and foundations
Probability and Statistics, Numerical Methods, Algorithms and Data Structures, and Languages, Automata and Computation.

AI and machine learning
Artificial Intelligence, Introduction to Machine Learning, Deep Learning Theory and Applications, and Computer Vision.

Systems
Operating Systems, Advanced Cloud Computing, and Database Management Systems.

Experience

Strategy Warsaw, Poland
AI Engineer May 2025 – Present

- Develop features for an enterprise agentic AI platform, including question-answer caching, dashboard recommendation workflows and conversational follow-up handling.
- Owned a customer-facing Gantt visualization from architecture and implementation through testing, cross-repository integration and QA coordination. Built a local test harness enabling iteration without rebuilding and deploying the full visualization library.
- Develop core functionality of an internal enterprise learning platform as a key contributor, using a spec-driven process and working with users, stakeholders and senior engineers on requirements and architecture.

Odra Labs Warsaw, Poland / Remote
Co-founder Oct 2025 – Aug 2026

- Built and deployed HermesFix, an agentic system for diagnosing and repairing WooCommerce issues using isolated execution, automated verification, human approval and reversible deployment.
- Prototyped Polis for controlled agent execution and monitoring, and MergeLearn, a local-first learning tool combining agent-authored lessons with spaced repetition.
- Contributed across architecture, implementation and product delivery. Secured AWS and Google startup credits.

Bank Gospodarstwa Krajowego Warsaw, Poland
Software Engineering Intern Jul 2023 – Sep 2023

- Tested banking-platform APIs and workflows with SoapUI and Jira. Reproduced and documented defects.

Technical Skills and Languages

Programming
Python for research and engineering. Academic projects in C++, Java, MATLAB and RISC-V assembly.

Languages
Polish (native), English (full professional proficiency).